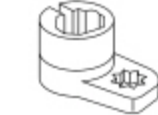


Engine Control Components — Exploded View, 3.5L GTDI

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST1447-A	Socket, Exhaust Gas Oxygen Sensor 303-476 (T94P-9472-A)
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

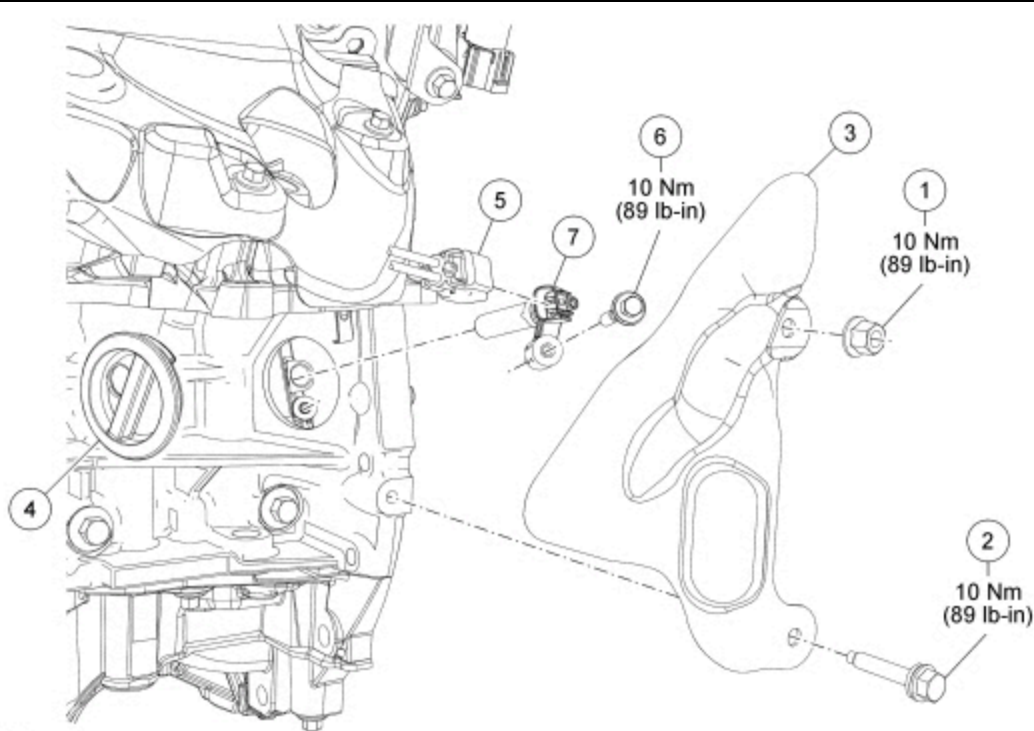
Material

Item	Specification
Motorcraft® High Temperature Nickel Anti-Seize Lubricant XL-2	—
Motorcraft® Penetrating and Lock Lubricant (US); Motorcraft® Penetrating Fluid (Canada) XL-1 (US); CXC-51-A (Canada)	—

Crankshaft Position (CKP) Sensor

NOTE: It is necessary to remove the LH turbo charger to access the Crankshaft Position (CKP) sensor. For additional information, refer to Section 303-04E.

NOTE: After installing the Crankshaft Position (CKP) sensor, use the scan tool to perform the Misfire Monitor Neutral Profile Correction procedure, following the on-screen instructions.

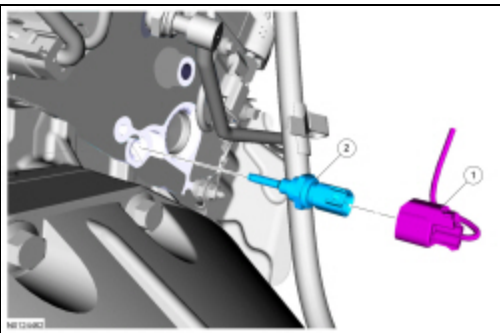


N0055318

Item	Part Number	Description
1	W701542 W701542	Heat shield nut
2	W701669 W701669	Heat shield bolt
3	6K342 6K342	Heat shield
4	6C070 6C070	Rubber grommet cover
5	—	Crankshaft Position (CKP) sensor electrical connector (part of 12C508)
6	W712315 W712315	CKP sensor bolt (part of 6D327)
7	6C315 6C315	CKP sensor (part of 6D327)

Cylinder Head Temperature (CHT) Sensor

NOTE: It is necessary to remove the RH fender splash shield to access the Cylinder Head Temperature (CHT) sensor. For additional information, refer to Section 501-02.



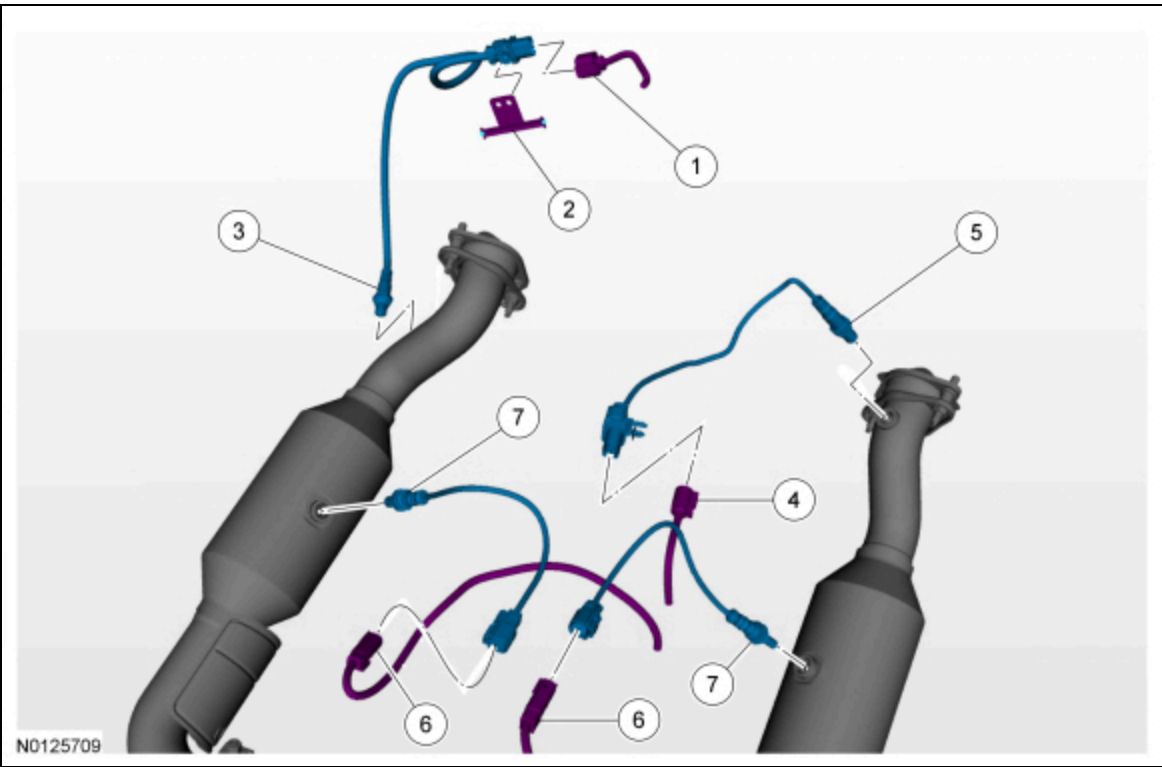
Item	Part Number	Description
1	—	Cylinder Head Temperature (CHT) sensor electrical connector (part of 12C508)
2	6G004 6G004	CHT sensor - 10 Nm (89 lb-In)

Heated Oxygen Sensor (HO2S) and Catalyst Monitor Sensor (CMS)

NOTE: If necessary, lubricate the Heated Oxygen Sensor (HO2S) and/or Catalyst Monitor Sensor (CMS) with penetrating and lock lubricant to assist in removal.

NOTE: Apply anti-seize to the threads of the HO2S and/or CMS prior to installing.

NOTE: The correct torque wrench setting must be calculated when using the Exhaust Gas Oxygen Sensor Socket to install the HO2S and/or CMS . [Refer to the Torque Wrench Adapter Formulas in the Appendix.](#)

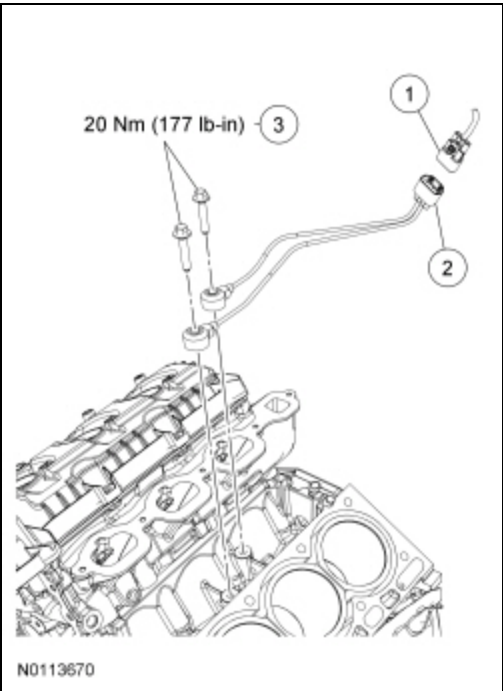


Item	Part Number	Description
1	—	Heated Oxygen Sensor (HO2S) electrical connector (part of 12C508)
2	—	HO2S wiring harness electrical connector retainer (part of 12C508)
3	9F4729F472	HO2S
4	—	HO2S electrical connector (part of 15525)
5	9F4729F472	HO2S
6	—	Catalyst Monitor Sensor (CMS) electrical connectors (part of 15525)
7	9G4449G444	CMS

Knock Sensor (KS)

NOTE: It is necessary to remove the fuel rail. For additional information, refer to Section 303-04C.

NOTE: LH cylinder head removed in graphic for clarity.



Item	Part Number	Description
1	—	Knock Sensor (KS) electrical connector (part of 12C508)
2	12A699 12A699	KS
3	W704749 W704749	KS bolt (2 required)

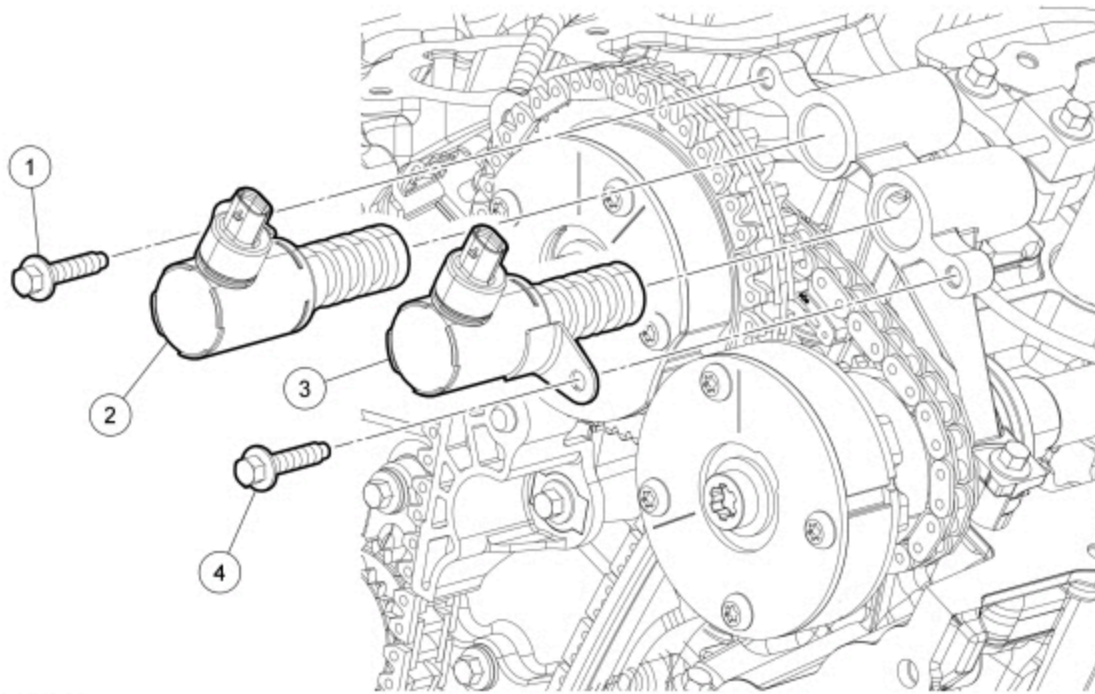
Variable Camshaft Timing (VCT) Oil Control Solenoid

NOTE: It is necessary to remove the RH or LH valve cover to access the Variable Camshaft Timing (VCT) oil control solenoid. For additional information, refer to Section 303-01A.

NOTE: Position the VCT oil control solenoid and tighten the bolts in 2 stages.

- Stage 1: Tighten to 8 Nm (71 lb-in).
- Stage 2: Tighten an additional 20 degrees.

NOTE: LH shown, RH similar.



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Item	Part Number	Description
1	W500215 W500215	LH intake Variable Camshaft Timing (VCT) oil control solenoid bolt
2	6B297 6B297	LH intake VCT oil control solenoid
3	6B297 6B297	LH exhaust VCT oil control solenoid
4	W500215 W500215	LH exhaust VCT oil control solenoid bolt

1. Refer to the procedures and/or exploded views in this section for any Warnings, Notices, Notes, Materials, Specifications, and Special Tools. Items in the exploded views may not be listed in order of removal.